Amazon Logistics & Delivery Performance Analysis (2025–2026)

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Tools Used:

- Excel
- SQL

Project Overview

Business Problem

Amazon needs to improve delivery efficiency and reduce delays while maintaining customer satisfaction.

Project Objectives

- Analyze delivery performance.
- Identify causes of late deliveries.
- Evaluate shipping methods.
- Assess hub performance.
- Examine regional delivery trends.
- Provide actionable recommendations.

Dataset Overview

- Order ID
- Order Date
- Shipping Method
- Delivery Status
- Hub
- Product Category
- Customer Rating
- Region

Tools Used

- SQL for data analysis
- Excel for data cleaning
- Excel for dashboard development

SQL Analysis

Query 1

Analyzed delivery performance by shipping method.

Amazon_Logi...\kkola (68))*

```sql
--Late Deliveries by Shipping

SELECT
SUM(CASE
WHEN Delivery_Status = 'Late Delivery' THEN 1
ELSE 0
END) Late_delivery,
Shipping_Method
FROM [Amazon Delivery Dataset.xlsx - Late Delivery Dataset]
GROUP BY Shipping_Method
ORDER BY Late_delivery DESC
```

123 % No issues found

Results Messages

	Late_delivery	Shipping_Method
1	114	Express Shipping
2	113	Same-Day Delivery
3	112	Two-Day Shipping
4	104	standard shipping

Query 2

Identified hubs contributing most to delays.

Amazon_Logi...\kkola (68))*

```sql
--Late delivery by warehouse

SELECT
Warehouse_Location,
SUM(CASE
WHEN Delivery_Status = 'Late Delivery' THEN 1
ELSE 0
END) Late_delivery
FROM [Amazon Delivery Dataset.xlsx - Late Delivery Dataset]
WHERE Warehouse_Location IS NOT NULL
GROUP BY Warehouse_Location
ORDER BY Late_delivery DESC
```

123 % No issues found

Results Messages

	Warehouse_Location	Late_delivery
1	Florida Hub	121
2	New York Hub	80
3	Texas Hub	80
4	California Hub	79
5	Illinois Hub	79

Query 3

Examined delivery performance by region.

```sql
--Late delivery by region

SELECT
Customer_State,
SUM(CASE WHEN Delivery_Status = 'Late Delivery' THEN 1
ELSE 0
END) Late_delivery
FROM [Amazon Delivery Dataset.xlsx - Late Delivery Dataset]
GROUP BY Customer_State
ORDER BY Late_delivery DESC

```

23 % No issues found

Results Messages Client Statistics

Customer_State	Late_delivery
Ohio	66
Texas	60
Nevada	58
California	57
Florida	56
Illinois	53
Georgia	50
New York	43

Query 4

Analyzed product categories with the highest delays.

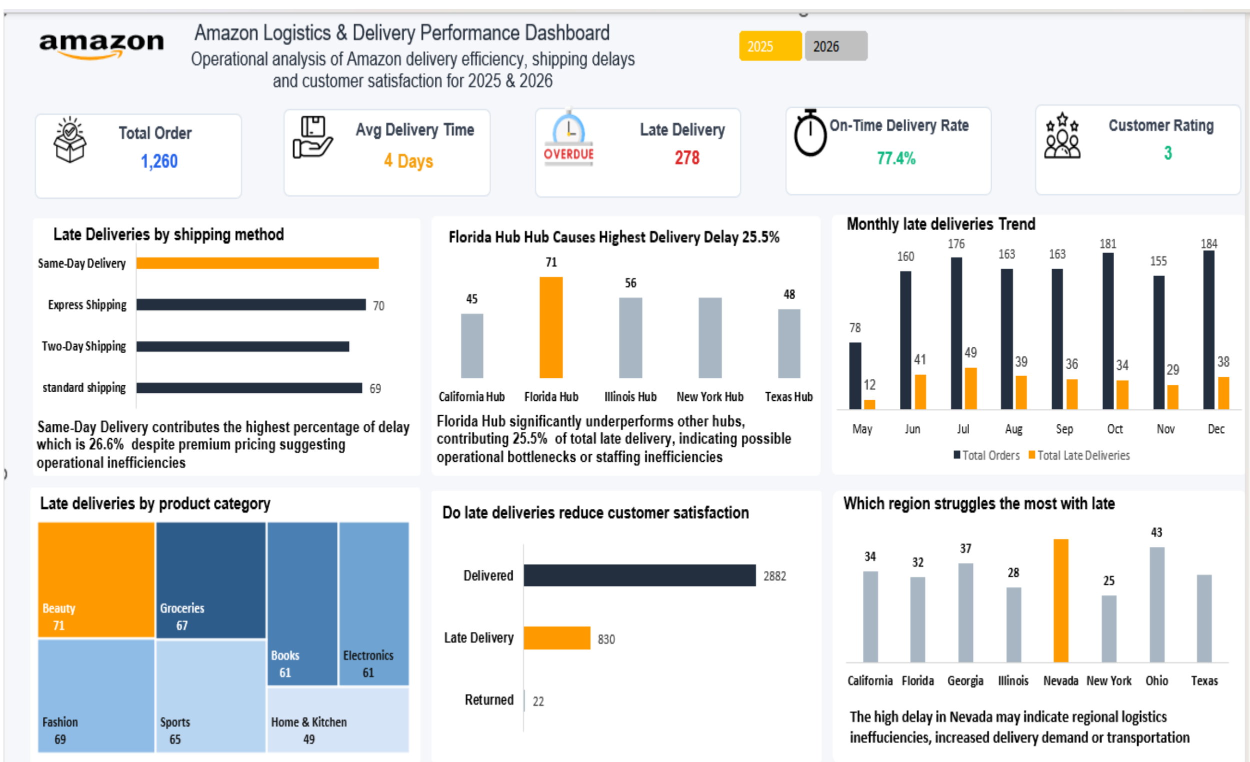
```sql
--Late Delivery by product category

SELECT
Product_Category,
SUM(CASE
WHEN Delivery_Status = 'Late Delivery' THEN 1
ELSE 0
END) Late_delivery
FROM [Amazon Delivery Dataset.xlsx - Late Delivery Dataset]
GROUP BY Product_Category
ORDER BY Late_delivery DESC
```

123 % No issues found

Results | Messages | Client Statistics

	Product_Category	Late_delivery
1	Beauty	71
2	Fashion	69
3	Groceries	67
4	Sports	65
5	Books	61
6	Electronics	61
7	Home & Kitchen	49

2025 Dashboard Analysis



Key Insights

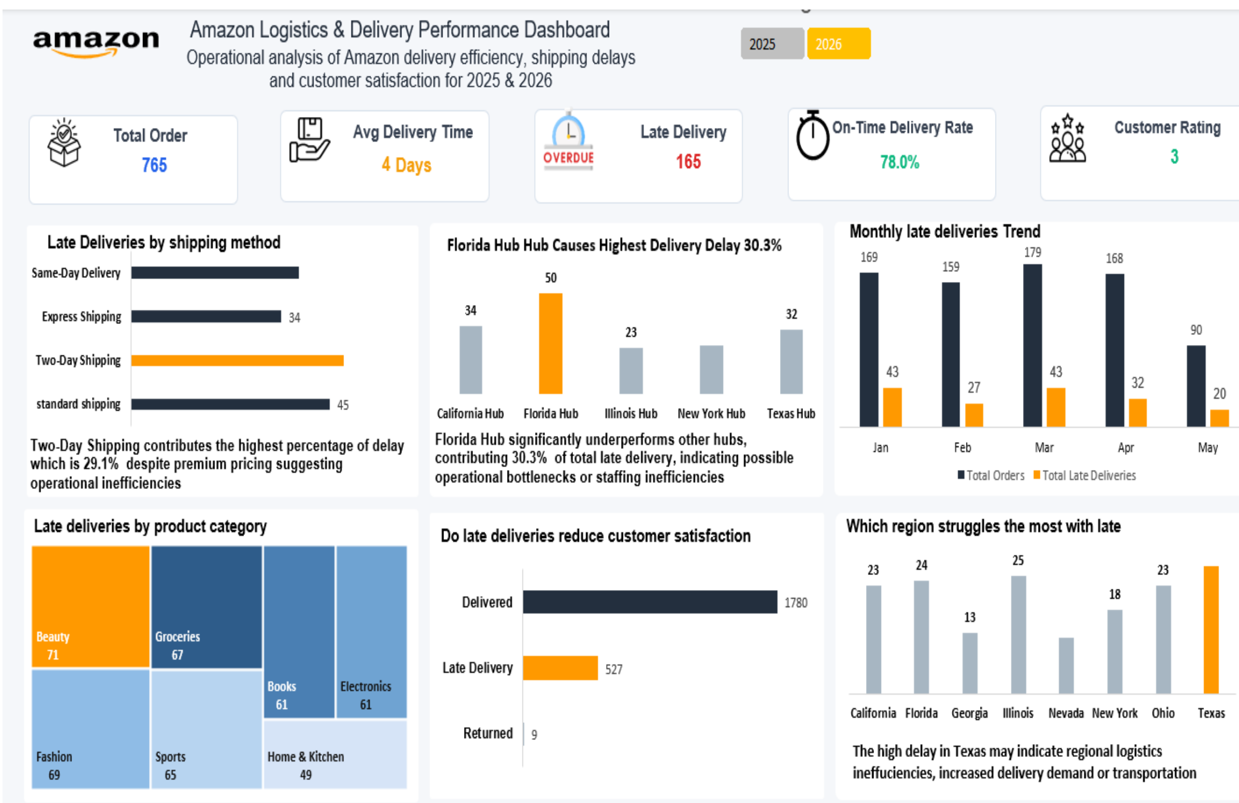
- 1,260 orders processed.
- On-time delivery rate was 77.4%.

- Same-Day Shipping recorded the highest delays.
- Florida Hub contributed 25.5% of delayed deliveries.
- Beauty products experienced the highest delays.
- Nevada recorded the highest delay volume.

Recommendations

- Improve Florida Hub operations.
- Review Same-Day delivery commitments.
- Strengthen inventory planning for Beauty products.
- Optimize delivery routes in Nevada.

2026 Dashboard Analysis



Key Insights

- 765 orders processed.
- On-time delivery rate improved to 78.0%.
- Late deliveries reduced to 165.

- Two-Day Shipping became the main source of delays.
- Florida Hub contributed 30.3% of delayed deliveries.
- Texas recorded the highest delay volume.

Recommendations

- Improve Florida Hub efficiency.
- Investigate Two-Day Shipping delays.
- Increase logistics resources in Texas.
- Improve demand forecasting.

Year-over-Year Comparison & Conclusion

Performance Comparison

Metrics	2025	2026
Orders	1,260	765
Late Deliveries	278	165
On-Time Rate	77.4%	78.0%
Main Delay Source	Same-Day	Two-Day
Top Delay Hub	Florida	Florida

Key Findings

- Delivery performance improved slightly in 2026.
- Florida Hub remained the primary operational bottleneck.
- Delay patterns shifted from Same-Day to Two-Day shipping.
- Customer satisfaction remained relatively unchanged.

Final Recommendations

- Prioritize Florida Hub optimization.
- Improve delivery planning during peak periods.
- Enhance shipping capacity management.
- Monitor customer satisfaction alongside delivery metrics.

Conclusion

This analysis evaluated Amazon's delivery performance across 2025 and 2026 using SQL, Excel, and Power BI. The findings revealed that while on-time delivery performance improved slightly and late deliveries decreased in 2026, operational challenges persisted, particularly at the Florida Hub, which remained the leading contributor to delivery delays. The analysis also identified shifts in delay patterns, with the primary source of delays moving from Same-Day Shipping in 2025 to Two-Day Shipping in 2026. By addressing hub inefficiencies, optimizing shipping operations, and improving regional logistics planning, Amazon can enhance delivery reliability, reduce delays, and improve overall customer satisfaction. This project demonstrates how data-driven insights can support strategic decision-making and operational improvement in logistics management.